Ohm's Law

Class 12 Science | Electricity | Concept, diagrams, solved examples and practice

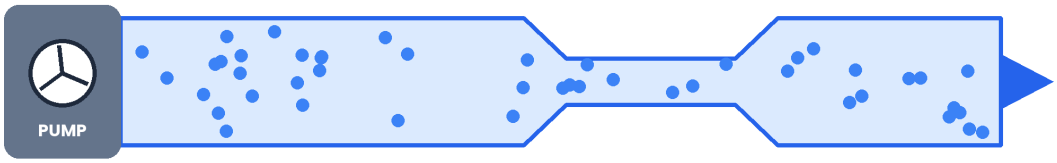
By the end of this chapter you can:

- state Ohm's law and write it as V = I R
- explain V, I and R using the water-in-a-pipe analogy
- verify the law with a circuit and draw the V–I graph
- solve numericals in board format, and use the V–I–R triangle as a quick check

1. The three quantities

Quantity	Symbol	Unit	Meaning
Potential difference	V	volt (V)	Work done per unit charge in moving charge between two points. 1 V = 1 J / 1 C.
Electric current	I	ampere (A)	Charge flowing per second, I = Q / t. 1 A = 1 C / 1 s.
Resistance	R	ohm (Ω)	How strongly a conductor opposes the flow of current. 1 Ω = 1 V / 1 A.

2. The water analogy



A pump pushes water through a pipe. The push is like **voltage**, the amount of water passing each second is like **current**, and a narrow section that makes the flow harder is like **resistance**. A stronger pump gives more flow through the same pipe.

3. Ohm's law

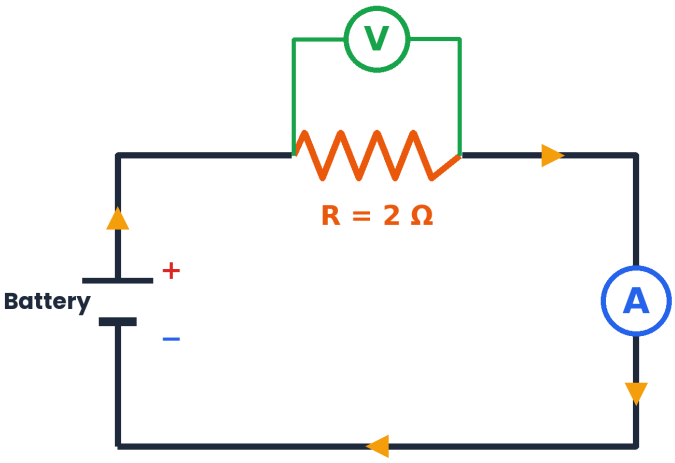
At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.

$$V \propto I \Rightarrow V = I R$$

R is a constant for that conductor. Rearranged: **I = V ÷ R** and **R = V ÷ I**.

4. Verifying Ohm's law in a circuit

Connect a battery, a resistor (2 Ω), an ammeter in series to measure the current, and a voltmeter in parallel across the resistor. Increase the voltage in steps and record the readings.



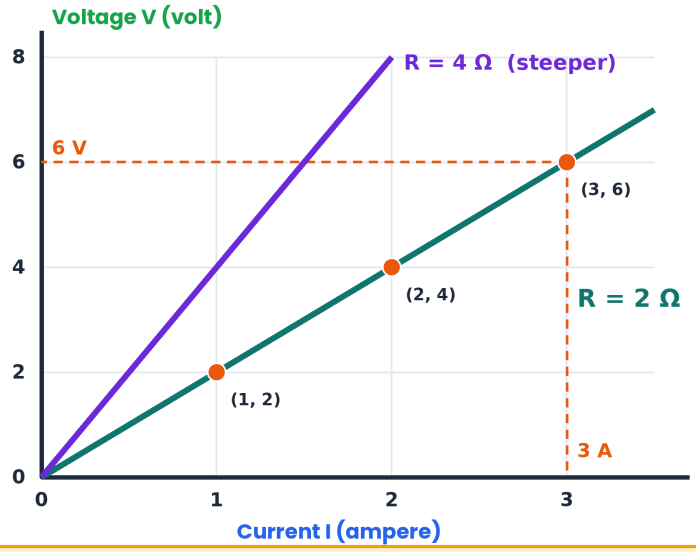
conventional current: from + terminal around the circuit

V (volt)	I (ampere)	V ÷ I (ohm)
2	1	2
4	2	2
6	3	2

Observation: when V doubles, I doubles; when V triples, I triples. The ratio V ÷ I stays the same (2 Ω). So V is directly proportional to I, and the constant ratio is the resistance R.

5. The V–I graph

Plot current on the x-axis and voltage on the y-axis. The points lie on a **straight line through the origin**. The **slope** of the line, V ÷ I, is the resistance. Here the slope is 6 ÷ 3 = 2 Ω. A steeper line means a larger resistance.



Note: Ohm's law holds for *ohmic* conductors at constant temperature, such as metallic wires. Devices such as diodes and filament lamps over a wide range do not follow it, but R = V ÷ I still gives the resistance at any one reading.

6. Solved examples

Solved example 1

Question	A bulb of resistance 20 Ω is connected to a 6 V battery. Find the current through it.
Given	R = 20 Ω, V = 6 V
Formula	V = I R ⇒ I = V ÷ R
Substitute	I = 6 ÷ 20
Answer	**I = 0.3 A**
Shortcut	6 ÷ 20 = 3 ÷ 10 = 0.3 A

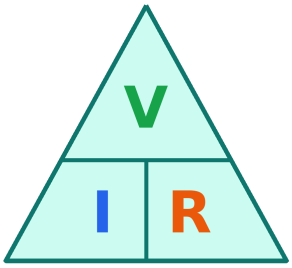
Solved example 2

Question	A heater draws a current of 5 A from a 230 V supply. Find its resistance.
Given	V = 230 V, I = 5 A
Formula	V = I R ⇒ R = V ÷ I
Substitute	R = 230 ÷ 5
Answer	**R = 46 Ω**
Shortcut	Cover R in the triangle: V over I = 230 ÷ 5 = 46 Ω

Solved example 3

Question	A current of 250 mA flows through a resistor of 8 Ω. Find the potential difference across it.
Given	I = 250 mA = 0.25 A, R = 8 Ω
Formula	V = I R
Substitute	V = 0.25 × 8
Answer	**V = 2 V**
Shortcut	250 mA = 1/4 A, so V = 8 ÷ 4 = 2 V (convert mA to A first)

7. Exam shortcut: the V–I–R triangle



Put **V** on top and **I** and **R** at the bottom. Cover the quantity you want:

- Cover **I**: I = V ÷ R
- Cover **V**: V = I × R
- Cover **R**: R = V ÷ I

Use it only for ohmic conductors at constant temperature. In a board answer, still write Given, Formula, Substitution and Answer with the unit.

8. Common mistakes

- Forgetting to convert mA to A (250 mA = 0.25 A) before using V = I R.
- Writing the answer without a unit, or with the wrong one (V for volt, A for ampere, Ω for ohm).
- Swapping the axes on the V–I graph: the slope equals R only when V is on the y-axis and I on the x-axis.
- Saying R changes with V or I. For an ohmic conductor R stays constant; it depends on material, length, thickness and temperature.

9. Practice questions

Write each answer in board format: Given, Formula, Substitution, Answer with unit.

Q1	A conductor has a resistance of 5 Ω. Find the current through it when the potential difference across it is 10 V.
Q2	A current of 0.5 A flows through a 12 Ω resistor. Find the potential difference across it.
Q3	A resistor draws 3 A from a 12 V battery. Find its resistance.
Q4	The V–I graph of a conductor is a straight line through the origin and passes through the point (2 A, 10 V). Find R.
Q5	A current of 500 mA flows through a resistor when 6 V is applied. Find the resistance.
Q6	If the potential difference across a fixed resistor is doubled, what happens to the current? Give a reason.

10. Answers

Q1	$I = V \div R = 10 \div 5 = 2$ A
Q2	$V = I\,R = 0.5 \times 12 = 6$ V
Q3	$R = V \div I = 12 \div 3 = 4$ Ω
Q4	$R = V \div I = 10 \div 2 = 5$ Ω (slope of the line)
Q5	500 mA = 0.5 A, $R = 6 \div 0.5 = 12$ Ω
Q6	The current also doubles, because $I = V \div R$ and R is constant ($V \propto I$).

Home Tuitions by Shagufa Khan

Notes, solutions and videos from Class 1 to NEET. Watch the video for this chapter and follow on Instagram for daily diagrams and questions.